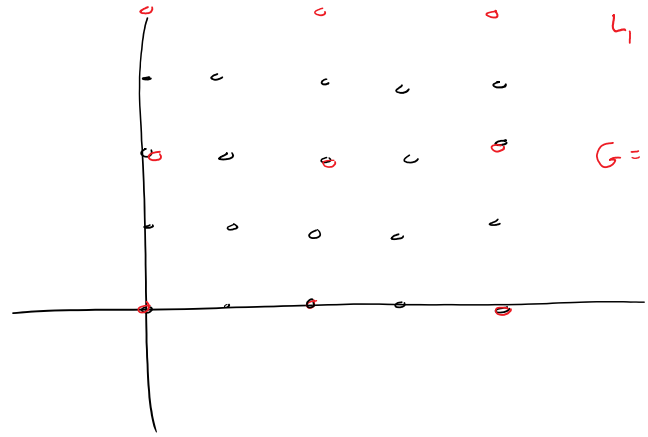


$L_1 \subset L_2$

$F = [b_1 \ b_2 \ \cdots \ b_n]$

$G = [g_1 \ g_2 \ \cdots \ g_n]$

$g_1 = F x_1$

$g_2 = F x_2$

$\vdots$

$g_n = F x_n$

$[g_1 \ g_2 \ \cdots \ g_n] = F \underbrace{[x_1 \ x_2 \ \cdots \ x_n]}_{X}$

$G = F X.$

$\underline{\det(G)} = \underline{\det(F)} \underbrace{\det(X)}_{\text{integer.}}$

$\det(F) \mid \det(G)$

---

$\begin{cases} G : L_1 \\ F : L_1 \end{cases}$

$G = F X_1$

$F = G X_2$

$\det(G) \mid \det(F)$

$\det(F) \mid \det(G)$

---

$x(t) \approx \sum_{i=1}^{n} c_i p_i(t)$

$X = \sum_{i=1}^{n} c_i P_i$

~~suppose $\{p_i\}$ are~~

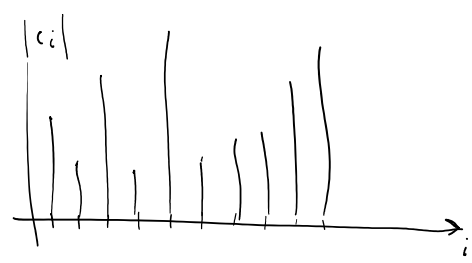
orthonormal,

$\boxed{c_i = \langle x(t), p_i(t) \rangle}$

Ex: $\underbrace{p_i(n) = e^{+jni\frac{2\pi}{N}}}$

$x(n) = \sum_{i=0}^{n-1} \underbrace{c_i} e^{+jni\, 2\pi/N}$

$c_i = \dfrac{\langle x(n), p_i(n) \rangle}{\langle p_i(n), p_i(n) \rangle}$



Error: $\left\| x(t) - \sum_{i=1}^{n} c_i p_i(t) \right\|_2^2$

$$= \left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ \underbrace{x(t) - \sum_{i=1}^{n} c_i p_i(t)}_{e(t)} \right\rangle$$

$$= \left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ x(t) \right\rangle - \underbrace{\left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ \sum_{i=1}^{n} c_i p_i(t) \right\rangle}_{0}$$

$$= \left\langle x(t), x(t) \right\rangle - \left\langle \sum_{i=1}^{n} c_i p_i(t), x(t) \right\rangle$$

$$= \left\langle x(t), x(t) \right\rangle - \left\langle \sum_{i=1}^{n} \langle x(t), p_i(t) \rangle p_i(t), x(t) \right\rangle$$

$$= \langle x(t), x(t) \rangle - \sum_{i=1}^{n} \underbrace{\langle x(t), p_i(t) \rangle}_{c_i} \underbrace{\langle p_i(t), x(t) \rangle}_{c_i^*}$$

$$= \|x(t)\|^2 - \sum_{i=1}^{n} |c_i|^2 \geq 0$$

$$\boxed{\|x(t)\|^2 \geq \sum_{i=1}^{n} |c_i|^2}$$ Bessel's inequality.

---

If $\{p_i(t)\}$ is complete, $x(t) = \sum_{i=1}^{n} c_i p_i(t)$

$$\left\| x(t) - \sum_{i=1}^{n} c_i p_i(t) \right\|^2 = 0$$

$$\Rightarrow \boxed{\|x(t)\|^2 = \sum_{i=1}^{n} |c_i|^2}$$ Parseval's thm.

---

$$x(t) = \sum_{i=1}^{n} c_i p_i(t)$$

$$y(t) = \sum_{i=1}^{n} d_i p_i(t)$$

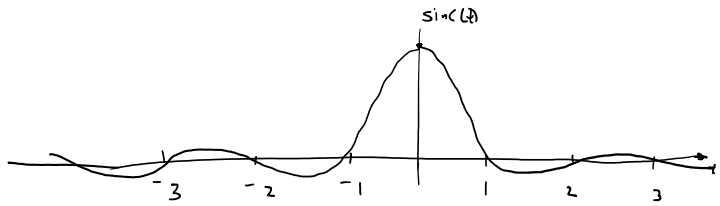
$$\boxed{\begin{aligned} \langle x(t), y(t) \rangle &= \sum_{i=1}^{n} c_i d_i^* \\ &= \langle \underline{c}, \underline{d} \rangle \end{aligned}}$$ Parseval's thm.

---

<u>sinc functions</u>

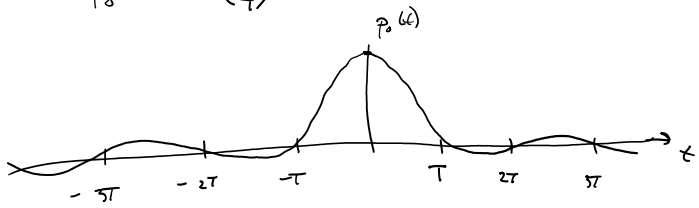
$$\text{sinc}(t) = \underline{\sin(\pi t)}$$

$\pi t$

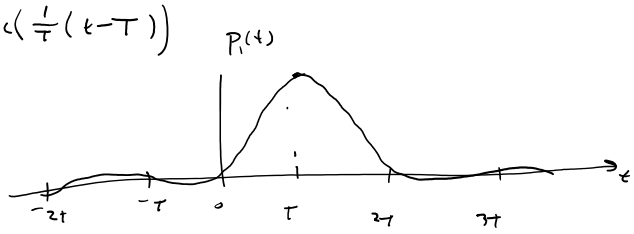



$$p_k(t) = \operatorname{sinc}\left(\frac{1}{T}(t - kT)\right)$$

$$p_0(t) = \operatorname{sinc}\left(\frac{t}{T}\right)$$




$$p_1(t) = \operatorname{sinc}\left(\frac{1}{T}(t - T)\right)$$

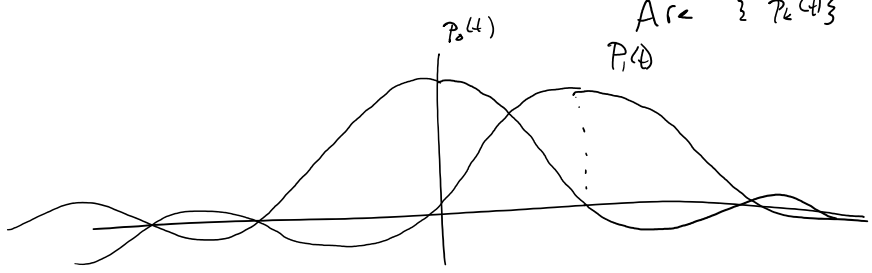



$p_0(t),\ p_1(t),\ \ldots$

$$x(t) = \sum_k c_k p_k(t)$$

How to find the coefficients?

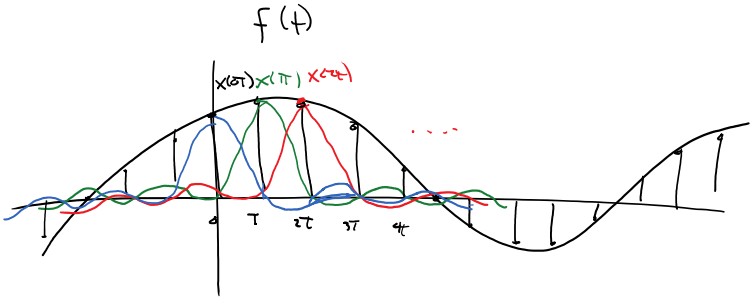
Are $\{p_k(t)\}$ orthogonal?




$$\int_{-\infty}^{\infty} p_i(t)\, p_j(t)\, dt = \int_{-\infty}^{\infty} \operatorname{sinc}\left(\frac{1}{T}(t - iT)\right) \operatorname{sinc}\left(\frac{1}{T}(t - jT)\right) dt$$

$$= 0 \quad \text{if } i \neq j.$$

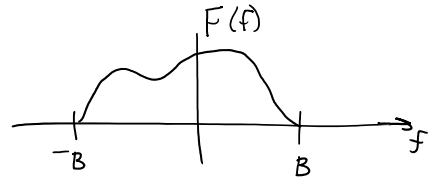
$$\boxed{c_k = \frac{\langle f(t), p_k(t) \rangle}{\langle p_k(t), p_k(t) \rangle}}$$




$$f(t) = \sum_{k=-\infty}^{\infty} x(kT)\, \mathrm{sinc}\left(\frac{1}{T}(t - kT)\right)$$

$f(t)$: B and limited.

$$f(t) \longleftrightarrow F(f)$$




sample interval $T = \frac{1}{2B}$

sample rate $= R = 2B$

---

Linear operators

Examples: $A\underline{x} = \underline{b}$

$A: \mathbb{R}^m \to \mathbb{R}^n$

$x \in \mathbb{R}^m \quad \underline{b} \in \mathbb{R}^n$

$A$: $n \times m$ matrix.

$$A(a x_1 + b x_2) = a A \underline{x}_1 + b A \underline{x}_2$$

Ex: $\underbrace{\frac{dx(t)}{dt} - a x = b(t)}$

$$\boxed{A x(t) = b(t)}$$

$$A = \frac{d}{dt} - a$$

$$A x(t) = \frac{d x(t)}{dt} - a x(t)$$

Ex: $\displaystyle\int_a^b k(t,\tau)\, x(\tau)\, d\tau = b(t)$

$$A x(t) = b(t)$$

$$A x(t) = \int_a^b \underbrace{k(t,\tau)}_{\text{kernel of the operator.}} x(\tau)\, d\tau$$

---

Linear Functional

$$f : X \longrightarrow \mathbb{R}$$

a set of functions. $L_2$

Ex: $f_1(x) = \frac{1}{T}\int_0^T x(t)\,dt$

$$f_2(x) = \int_a^b x(t)\,\phi(t)\,dt$$

$$f_3(x) = \int_a^b x(t)\,e^{-jwt}\,dt$$

Fact: If $X$ is a Hilbert space, then any continuous functional can be expressed as an inner product.

Riesz Representation thm.

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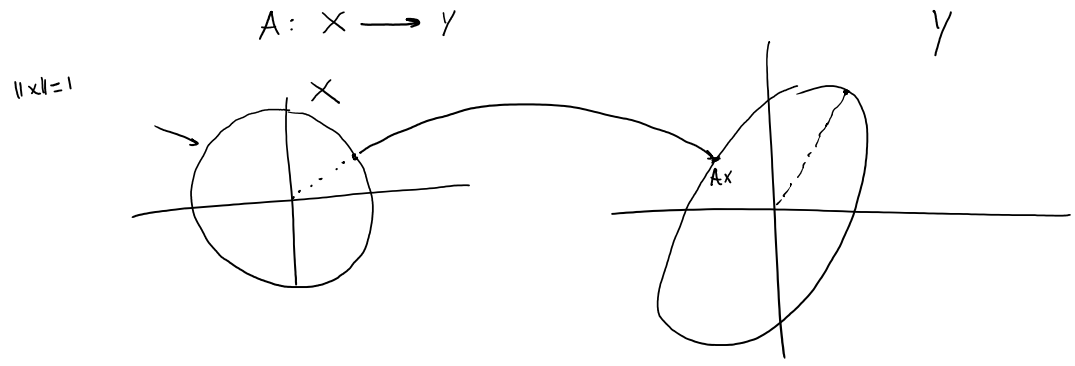
## Operator Norms

$$A : X \longrightarrow Y$$




$$\|A\|_p = \sup_{x:\|x\|_p=1} \|Ax\|_p$$

$$= \sup_{x:x\neq 0} \frac{\|Ax\|_p}{\|x\|_p}$$

---

submultiplicative property.

$$\|AB\|_p \le \|A\|_p \, \|B\|_p.$$

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Bounded operators